

MODDABLE CHESS · VARIANT 98 OF 155

# Möbius Strip Chess

**8×8 Möbius strip**

BOARD

**2**

PLAYERS

# MöBIUS STRIP CHESS

Möbius Strip Chess is chess played on a **Möbius strip** — a surface with a single side and a single boundary edge, created by joining one pair of opposite edges with a half-twist.

## THE TOPOLOGY

**Free edges:** The a-file and h-file are genuine boundaries. Pieces cannot move beyond these files — there is no horizontal wrapping. These form the single boundary edge of the Möbius strip.

**Twisted seam:** Rank 1 and rank 8 connect with a half-twist. The file mapping is identical to Klein Bottle Chess:

| FILE LEAVING AT RANK 8 | FILE ARRIVING AT RANK 1 |
|------------------------|-------------------------|
| a                      | h                       |
| b                      | g                       |
| c                      | f                       |
| d                      | e                       |
| e                      | d                       |
| f                      | c                       |
| g                      | b                       |
| h                      | a                       |

A piece moving off rank 8 on file d arrives on rank 1 on file e. Movement direction is preserved; file position is mirrored.

**One side:** A piece that crosses the seam once is on the “other side” of the strip. Crossing the seam a second time returns it to the original side. A full circuit of the board (two seam crossings) returns the piece to its starting square.

## CONSEQUENCES FOR PIECES

- ♦ **Rooks:** Can slide off rank 8 and continue from rank 1 on a mirrored file. A Rook that slides forward and crosses the seam is now moving towards its starting side
- ♦ **Bishops:** Change colour when crossing the seam (non-orientability reverses the diagonal colouring). A bishop on a light square that crosses the seam lands on a dark square
- ♦ **Queens:** Combined horizontal (blocked at a/h-file) and vertical (twisted seam) movement
- ♦ **Sliders crossing the boundary:** Cannot slide beyond the a-file or h-file — those are genuine walls
- ♦ **Knights:** Leaps that would land beyond the a/h boundary are illegal. Leaps across the rank seam land on the mirrored file

## STARTING POSITION AND RULES

Standard FIDE starting position. All FIDE rules apply; movements respect the free edges and twisted rank seam.

Win: Checkmate the opponent's King.

## ATTRIBUTION

Möbius Strip Chess is a documented chess variant applying non-orientable surface topology to FIDE chess.

Source: [chessvariants.com](http://chessvariants.com)